Lindenbaum lemma: every Sigma-consistent set extends to a complete Sigma-consistent one

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof's parenthetical on transitivity of the stage inclusions is verbatim; the enumeration recipe (variables p_0..p_n by length) is the author's explicit example.

Editorial note. The label thm:lindenbaum names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own thm environment in the unit source.

Allowed standard prerequisites (editorial).

- Enumerability of formulas. Same part's setup: formulas can be listed A_0, A_1, ... by length over growing variable pools. Lines 1033-1036: the explicit listing recipe opening the proof.
- Consistency facts: for each A, at least one of Delta+{A}, Delta+{not A} stays consistent. Proof theory part, prop:consistencyfacts item (c) (cited by double olref). Line 1087: the case split inside the inductive consistency preservation.
- Finitude of derivations (compactness of the turnstile). Definition of Sigma-derivability via a single conditional over finitely many premises. Lines 1067-1075: an inconsistency witness involves finitely many Delta-members, hence lives at some finite stage.

Source reference. Original proposition prop-consistencyfacts: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1088–1088. Named citation invoked at source line(s) 1088: the author's marker \olref[prf][con]{prop:consistencyfacts} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-consistencyfacts-c: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1088–1088. Named citation invoked at source line(s) 1088: the author's marker \olref[prf][con]{prop:consistencyfacts-c} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1 (Lindenbaum's Lemma)**.** *If* Γ *is* Σ*-consistent then there is a complete* Σ*-consistent set* Δ *extending* Γ*.*

Proof. Let A_0, A_1, … be an exhaustive listing of all formulas of the language (repetitions are allowed). For instance, start by listing p_0, and at each stage $n \geq 1$ list the finitely many formulas of length n using only variables among p_0, …, p_n. We define sets of formulas Δ_n by induction on n, and we then set $\Delta = \bigcup_n \Delta_n$. We first put $\Delta_0 = \Gamma$. Supposing that Δ_n has been defined, we define Δ_{n+1} by:

$$\Delta_{n+1} = \begin{cases} \Delta_n \cup \{A_n\}, & \text{if } \Delta_n \cup \{A_n\} \text{ is } \Sigma\text{-consistent;} \\ \Delta_n \cup \{\neg A_n\}, & \text{otherwise.} \end{cases}$$

Now let $\Delta = \bigcup_{n=0}^{\infty} \Delta_n$.

We have to show that this definition actually yields a set Δ with the required properties, i.e., $\Gamma \subseteq \Delta$ and Δ is complete Σ-consistent.

It's obvious that $\Gamma \subseteq \Delta$, since $\Delta_0 \subseteq \Delta$ by construction, and $\Delta_0 = \Gamma$. In fact, $\Delta_n \subseteq \Delta$ for all n, since Δ is the union of all Δ_n. (Since in each step of the construction, we add formula to the set already constructed, $\Delta_n \subseteq \Delta_{n+1}$, so since $\subseteq$ is transitive, $\Delta_n \subseteq \Delta_m$ whenever $n \leq m$.) At each stage of the construction, we either add A_n or $\neg A_n$, and every formula appears (at least once) in the list of all A_n. So, for every A either $A \in \Delta$ or $\neg A \in \Delta$, so Δ is complete by definition.

Finally, we have to show, that Δ is Σ-consistent. To do this, we show that (a) if Δ were Σ-inconsistent, then some Δ_n would be Σ-inconsistent, and (b) all Δ_n are Σ-consistent.

So suppose Δ were Σ-inconsistent. Then $\Delta \vdash [\Sigma]\bot$, i.e., there are A_1, …, $A_k \in \Delta$ such that $\Sigma \vdash A_1 \to (A_2 \to \cdots (A_k \to \bot) \dots)$. Since $\Delta = \bigcup_{n=0}^{\infty} \Delta_n$, each $A_i \in \Delta_{n_i}$ for some n_i. Let n be the largest of these. Since $n_i \leq n$, $\Delta_{n_i} \subseteq \Delta_n$. So, all A_i are in some Δ_n. This would mean $\Delta_n \vdash [\Sigma]\bot$, i.e., Δ_n is Σ-inconsistent.

To show that each Δ_n is Σ-consistent, we use a simple induction on n. $\Delta_0 = \Gamma$, and we assumed Γ was Σ-consistent. So the claim holds for $n = 0$. Now suppose it holds for n, i.e., Δ_n is Σ-consistent. Δ_{n+1} is either $\Delta_n \cup \{A_n\}$ if that is Σ-consistent, otherwise it is $\Delta_n \cup \{\neg A_n\}$. In the first case, Δ_{n+1} is clearly Σ-consistent. However, by prop-consistencyfactsprop-consistencyfacts-c, either $\Delta_n \cup \{A_n\}$ or $\Delta_n \cup \{\neg A_n\}$ is consistent, so Δ_{n+1} is consistent in the other case as well. □